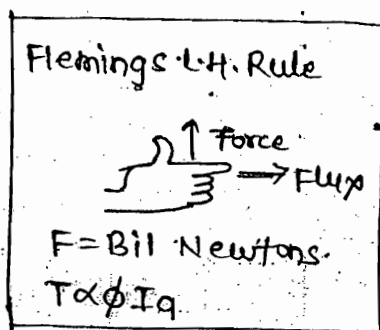
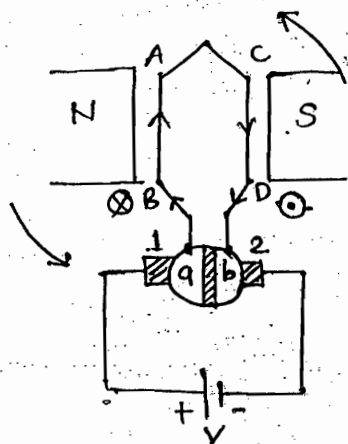


## DC MOTORS

- \* These are more popular because of their highest starting torque & wide range of accurate, simple & efficient speed control.
- \* The same d.c. gen<sup>r</sup> can be operated as a motor.



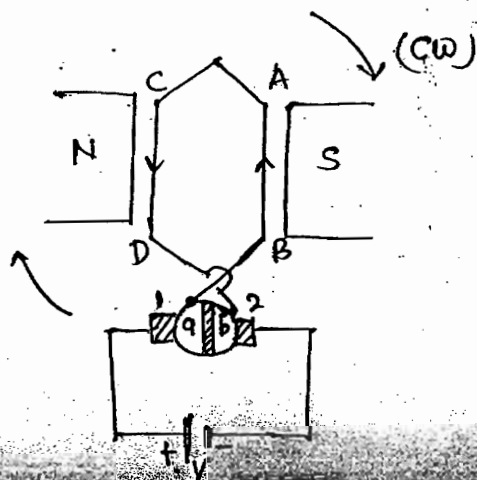
Principle →

- \* When a current carrying cond<sup>r</sup> placed in a magnetic field it will experience a mech. force which magnitude is given by

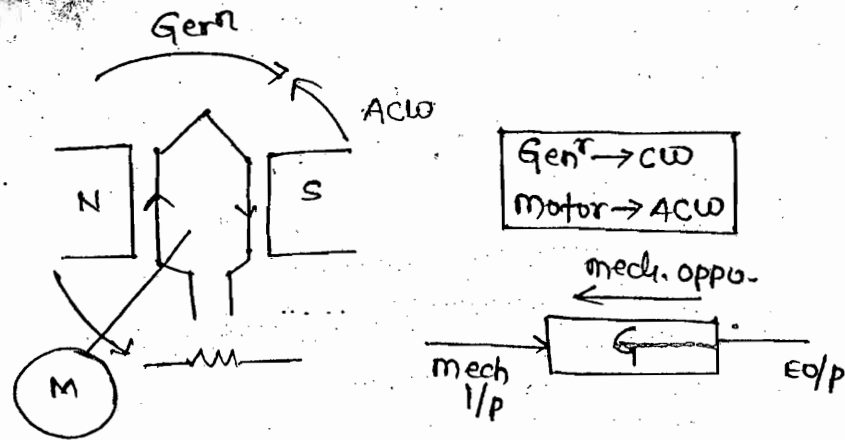
$$F = B i l \text{ Newtons}$$

& the dir<sup>n</sup> acc to FHL Rule also called as motor Rule. For the same notation in case of a gen<sup>r</sup>. If it is rotated clockwise the motor will rotate anticlockwise.

- \* The fun<sup>n</sup> of commutator is to produce unidirectional torque.

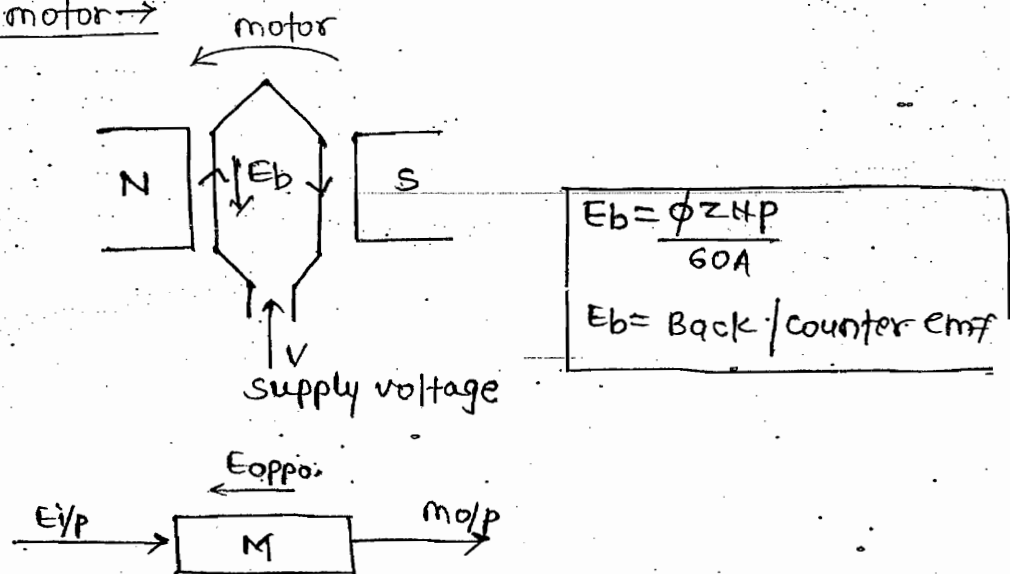


## Motor Action in a Genr



- \* Energy Conversion from one form to other occurs through an opposition acc. to basic fundamental laws of nature.
- \* In a genr mech. to ele. energy happens through mech. opposition known as motor action in genr.
- \* The prime mover which is rotating the genr should be capable of rotating the genr against the backward force or torque.

## Genr Action in motor



- \* In the motor arm. Cond<sup>r</sup> when they rotate voltage is induced which is acc. to Fleming's Right hand Rule & the induced emf is

$$E_g = \frac{\phi ZNP}{60A}$$

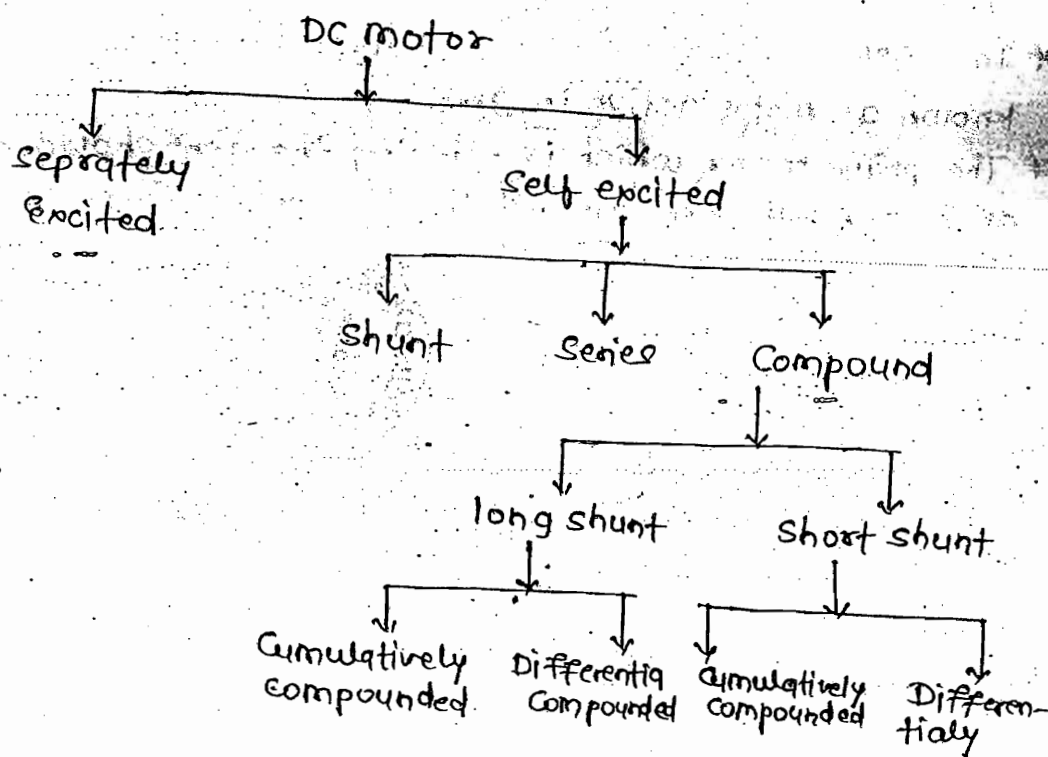
\* By applying right hand rule in motor the dirn of induced emf is exactly opposite to supply vol.

Therefore it is known as back emf or counter emf.

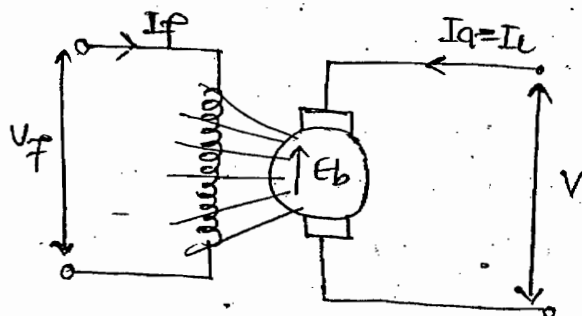
$$E_b = \frac{\phi ZNP}{60A}$$

$$E_b \propto \phi N$$

\* The construction of dc motors are identical to gen<sup>r</sup> therefore the classification is similar.



separately excited →

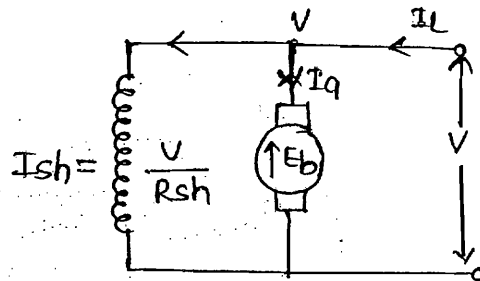


$$V = E_b + I_a R_a + BCD$$

$$I_a = I_L$$

\* It basically used in servo motor.

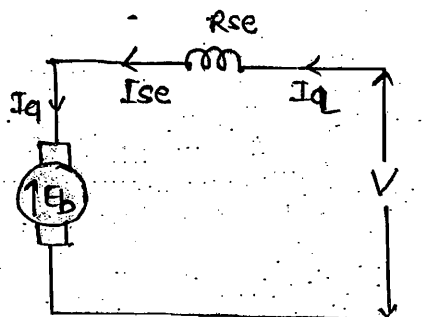
### ② Shunt motor →



$$V = E_b + I_a R_a + BCD$$

$$I_L = I_a + I_{sh}$$

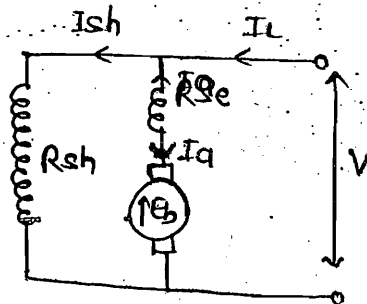
### ③ Series Motor →



$$V = E_b + I_a (R_a + R_{se}) + BCD$$

$$I_a = I_{se} = I_L$$

### ④ Long shunt →

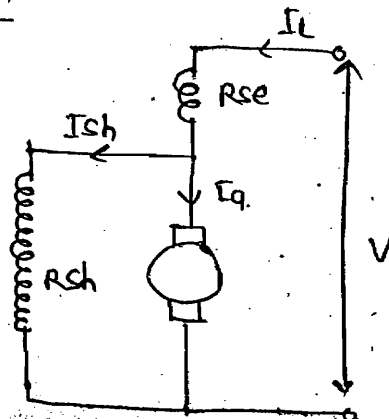


$$V = E_b + I_a (R_{se} + R_a) + BCD$$

$$I_L = I_a + I_{sh}$$

$$I_{sh} = V / R_{sh}$$

### ⑤ Short shunt →

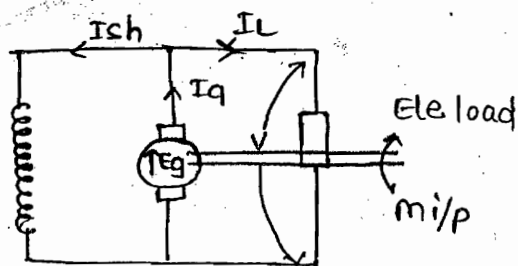


$$V = E_b + I_L R_{se} + I_a R_a + BCD$$

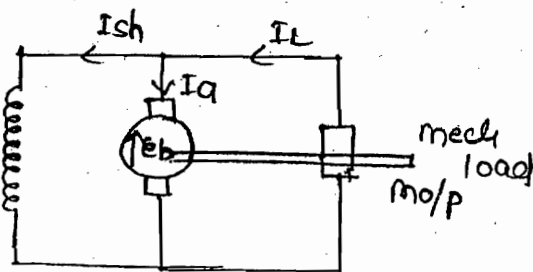
$$I_L = I_{sh} + I_a$$

$$I_{sh} = \frac{V - I_L R_{se}}{R_{sh}}$$

### Shunt Genr



### Shunt motor



(1) Genr delivers

$$I_a = I_L + I_{sh}$$

(2)  $E_g > V$

(3)  $E_g \rightarrow$  Gen/Ind. emf

(4)  $V \rightarrow$  Terminal vol.

(5)  $E_g I_a \rightarrow$  Ele. power generated in arm.

(6)  $V I_L \rightarrow$  Power delivered to the load.

(7)  $E_g I_a - V I_L =$  Total cu loss

(8) Electrical loading

(1) Motor draws

$$I_L = I_a + I_{sh}$$

(2)  $E_b < V$

(3)  $E_b \rightarrow$  Induced/Gen Back emf

(4)  $V$  : Supply vol.

(5)  $E_b I_a$  : - Ele. eq. of mech. power developed in arm.

(6)  $V \cdot I_L \rightarrow$  Ele. power i/p

(7)  $V \cdot I_L - E_b I_a \rightarrow$  Total cu loss

(8) Mechanical loading

### \* Significance of Back emf $\rightarrow$

(1) It is playing a role of opposition in electromech. energy conversion in motor.

(2) The  $\eta$  of motor is depending on its back emf (directly proportional)

$$V = E_b + I_a R_a$$

$$V I_a = E_b I_a + I_a^2 R_a$$

$$i/p = o/p + loss$$

$$\eta = \frac{o/p}{i/p} = \frac{E_b I_a}{V I_a} = \frac{E_b}{V}$$

$$\boxed{\eta = \frac{E_b}{V}}$$

③ Mech. power developed :-

$$\text{max}^m \text{ Cond} \rightarrow P_m = V I_a - I_a^2 R_a$$

$$\frac{dP_m}{dt} \Rightarrow 0 = V - 2 I_a R_a$$

$$\boxed{I_a R_a = V/2}$$

If  $I_a R_a$  (drop) is  $V/2$  then the

$$V = E_b + I_a R_a$$

$$V = E_b + V/2$$

$$\boxed{E_b = V/2}$$

Now when (m)  $P_m$  max then (m)

$$\eta = \frac{E_b}{V} = \frac{V/2}{V} = 1/2 = 50\%$$

$$\boxed{\eta = 50\%}$$

Now:

$$V = E_b + I_a R_a$$

$$I_a = \frac{V - E_b}{R_a} = \frac{V - V/2}{R_a} = \frac{V}{2 R_a}$$

$$\boxed{I_a = \frac{V}{2 R_a}}$$

Here value of current increases ( $I_a \uparrow$ ).

\* The mech. power developed will be max<sup>m</sup> if the back emf is half the supply vol.

\* Under such cond<sup>n</sup> the  $\eta$  of the motor is only 50%. Therefore dc motors are not designed for such cond<sup>n</sup> but they are designed to give max<sup>m</sup>  $\eta$  near rated load cond<sup>n</sup>.

(4) Back emf makes the motor self regulating in nature.

\* When the motor is loaded it has to develop the electromagnetic torque which equals the load torque.

\* It will do the same by reducing its speed, back emf & increasing the current proportionally.

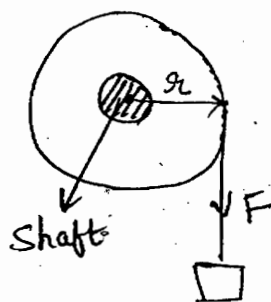
\* When the electromagnetic torque developed is exactly equal to the load torque the speed will not reduce further & the motor run at new speed.

\* If the load is removed the speed increase but do not rise because the back emf will regulate the current & torque.

$$I_a = \frac{V - E_b}{R_a}$$

\* TORQUE → \* It is the turning / twisting movement of force about an axis expressed in N-m.

$$T = F \times r \text{ N-m}$$



$$P = \frac{\text{Work done}}{\text{time}} = \frac{\text{Force} \times \text{distance}}{\text{time}}$$

$$P = \frac{2\pi r \times F}{60/N}$$

$$P = \frac{2\pi N (F \cdot r)}{60}$$

$$P = \frac{2\pi NT}{60} \text{ watts}$$

Power developed in the arm.

$$P = E_b \cdot I_a$$

$$E_b I_a = \frac{2\pi NT}{60}$$

$$I_a \frac{\phi Z P}{60 A} = \frac{2\pi NT}{60}$$

$$T = \frac{\phi Z P \cdot I_a}{2\pi A}$$

$$T = \frac{1}{2\pi} \cdot \frac{\phi Z P I_a}{A}$$

$$T = \frac{1}{2\pi} \times \phi I_a \times \frac{Z P}{A}$$

$$T \propto \phi I_a$$

$$\frac{T_2}{T_1} = \frac{\phi_2}{\phi_1} \times \frac{I_{a2}}{I_{a1}}$$

Flux constant:-

$$\frac{T_2}{T_1} = \frac{I_{a2}}{I_{a1}}$$

If flux constant shunt motors.

$$\frac{T_2}{T_1} = \left( \frac{I_{a2}}{I_{a1}} \right)^2$$

For series motor upto saturation  
( $I_a \propto \phi_{se}$ )

After saturation again shunt motor

$$\frac{T_2}{T_1} = \frac{I_{a2}}{I_{a1}}$$

$$E_b \propto N \phi$$

$$N \propto \frac{E_b}{\phi}$$

$$N \propto \frac{V - I_a R_a}{\phi}$$

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$$

If flux constant.

Armature torque:-

$$\frac{2\pi N T_a}{60} = P = E_b I_a = \text{Rotational losses}$$

↓  
IRON & mech.

$$T_a - T_{sh} = \text{lost torque}$$

Characteristics →

(1) Torque vs  $I_a$

(2) Speed vs  $I_a$

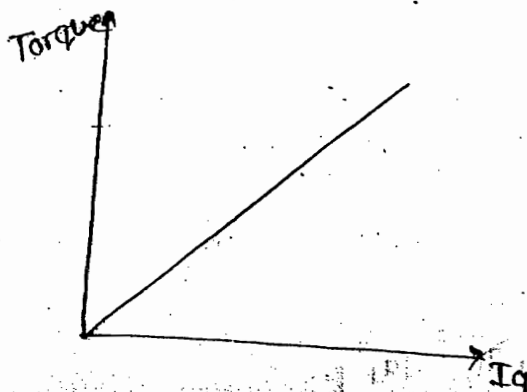
(3) Speed vs  $T$

} same

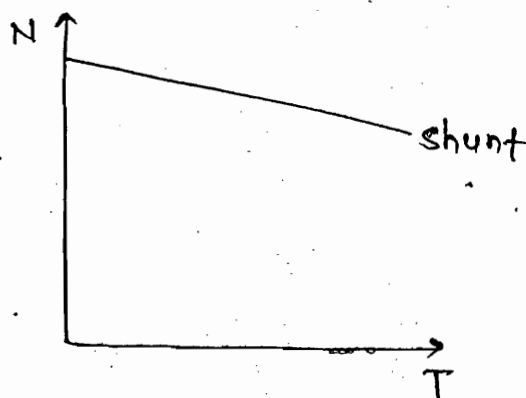
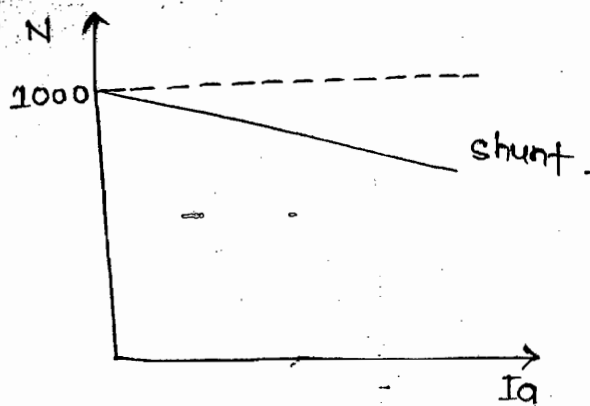
① shunt motor →

① Torque vs  $I_a$

$$T \propto I_a$$

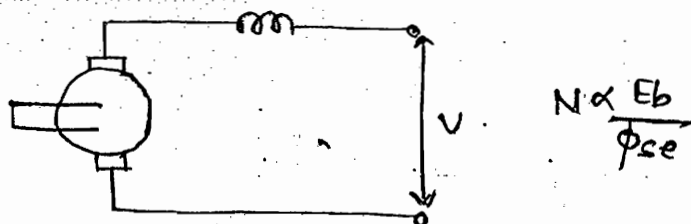


$$N \propto V - I_a R_a$$



- \* Shunt motors have excellent speed c/s. & they are approx. as constant speed motors.
- \* They have low or medium starting torque.
- \* These are best suitable for manufacturing purposes like steel rolling, Al rolling, lathe's, m/c tools.
- \* The c/s of shunt motors are identical to ind<sup>n</sup> motors. but shunt motors are preferred for there superior speed control.

(2) Series Motors →



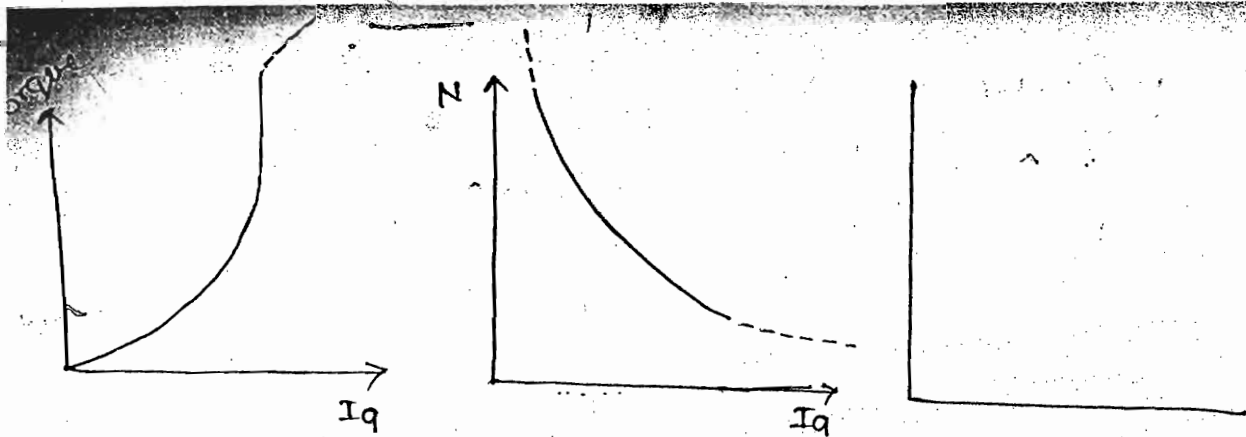
At NL the  $I_a \downarrow$  then  $\phi_{se} \downarrow$  & speed  $\uparrow$ .

$$I_a = \frac{V - E_b}{R_a}$$

$I_a \downarrow$   $E_b$  will max<sup>m</sup>

$$\uparrow E_b \propto \phi_{se} \downarrow N \uparrow$$

Never start series motor at NL because speed becomes dangerous.



$$T \propto \phi_s I_a$$

$T \propto I_a^2$  Before Saty.

$T \propto I_a$  After saturation.

$$N \propto \frac{V - I_a R_a}{\phi_s}$$

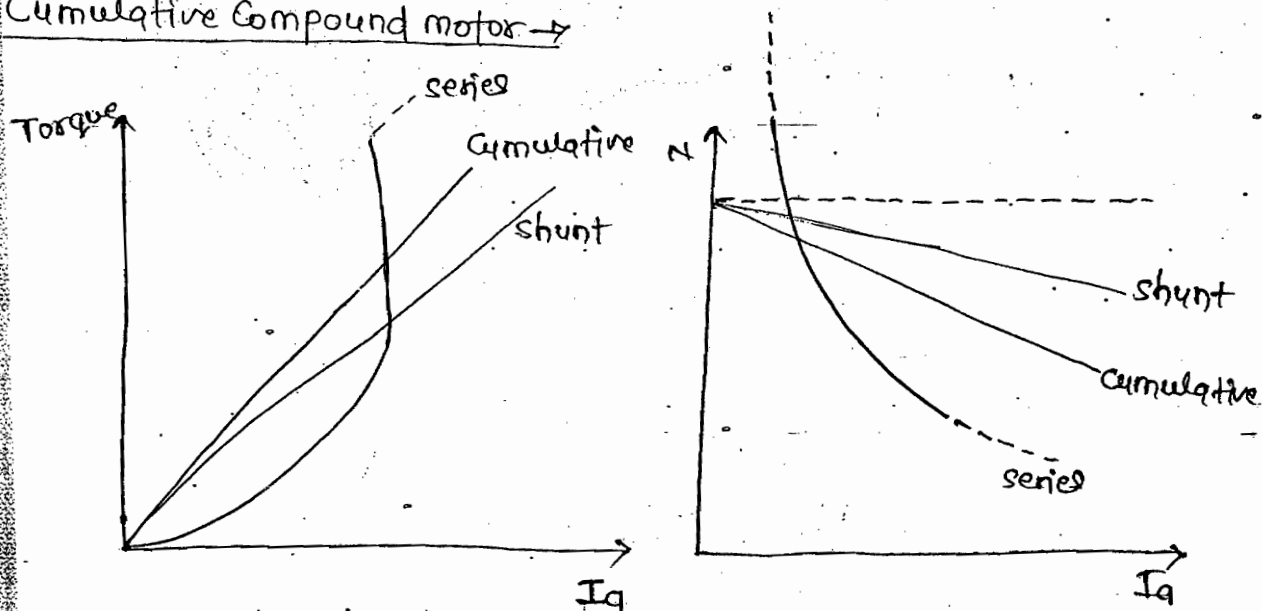
\* The series motors have highest starting torque. Therefore used in electric traction, cranes, hoists etc.

\* These are not suitable for belt drives.

\* Never start a series motor at NL its speed becomes dangerously high & motor damages mechanically.

\* They have variation in speed with load due to series flux & dangerous speed on NL.

Cumulative Compound motor →



$$T \propto (\phi_{sh} + \phi_s) I_a$$

\* It is best suitable for high torque intermittent load cond<sup>n</sup>

like shears & punches, presses, compressors.

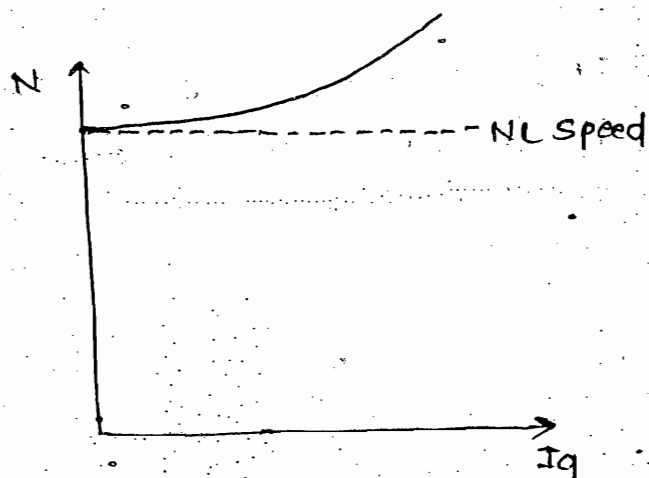
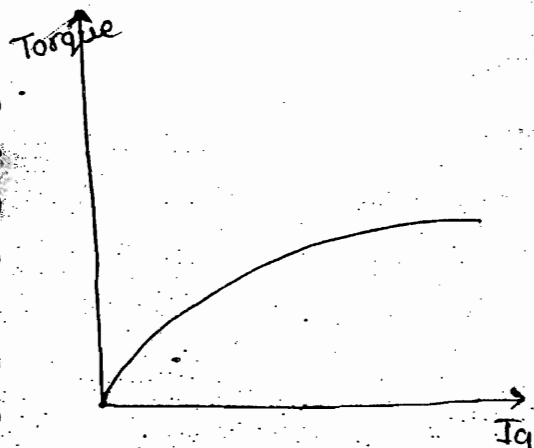
\* They have definite NL speed.

\* The shunt flux regulates the speed on NL cond<sup>n</sup> & the series flux increases the torque with load cond<sup>n</sup>

(4) Differentially Compounded →

$$T \propto (\phi_{sh} - \phi_{se}) I_a$$

$$I_a \uparrow \quad \phi_{se} \uparrow \quad (\phi_{sh} - \phi_{se}) \downarrow \quad T \downarrow$$



\* Due to unstable torque & speed c/s they don't have practical app<sup>n</sup>

\* Speed Regulation →

\* The change in speed when the Full load across the motor is disconnected expressed in % of rated speed.

$$\%SR = \frac{N_0 - N}{N} \times 100$$

\* shunt motors have best SR.

\* Differential motors have -ve SR. & these motors may have 0 speed reg<sup>n</sup> without any external control.

(26/6)

$$T \propto \phi_{sh} I_a \text{ (shunt)}$$

$$T \propto \phi_{se} I_a \text{ (series)}$$

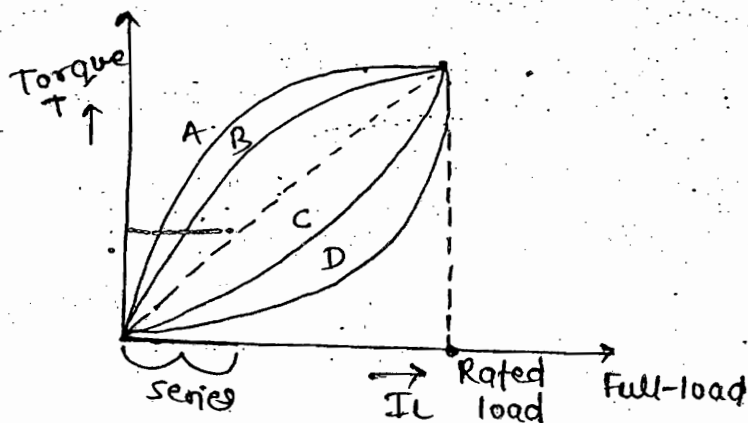
$$T \propto (\phi_{sh} + \phi_{se}) I_a \text{ (cumu.)}$$

$$T \propto (\phi_{sh} - \phi_{se}) I_a \text{ (Diff.)} \quad I_a \uparrow \phi_{se} \uparrow (\phi_{sh} - \phi_{se}) \downarrow$$

Ans. (d)

\*\* During starting diff motors may run in the reverse dir<sup>n</sup> if the series flux dominates shunt flux due to large starting current flowing through arm.

(29/6)



At Rated load each motor give same c/s

$$T \propto \phi_{sh}$$

$$T \propto \phi_{se}$$

$$T \propto (\phi_{sh} + \phi_{se})$$

$$T \propto (\phi_{sh} - \phi_{se})$$

(Because  $I_a$  same  
No comparison)

D  $\rightarrow$  Series motor (Because in series motor negligible value of flux at NL)